

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	19 July 2026
Team ID	
Project Name	SB Stocks (Stock Trading App)
Maximum Marks	4 Marks

Technical Architecture:

The architectural solution is as depicted in the architectural diagram as below and the information as per the table 1 & table 2.

Example: Stock trading web application for paper trading and portfolio management.

Reference: <https://github.com/issak-04/STOCK-TRADING-APP>

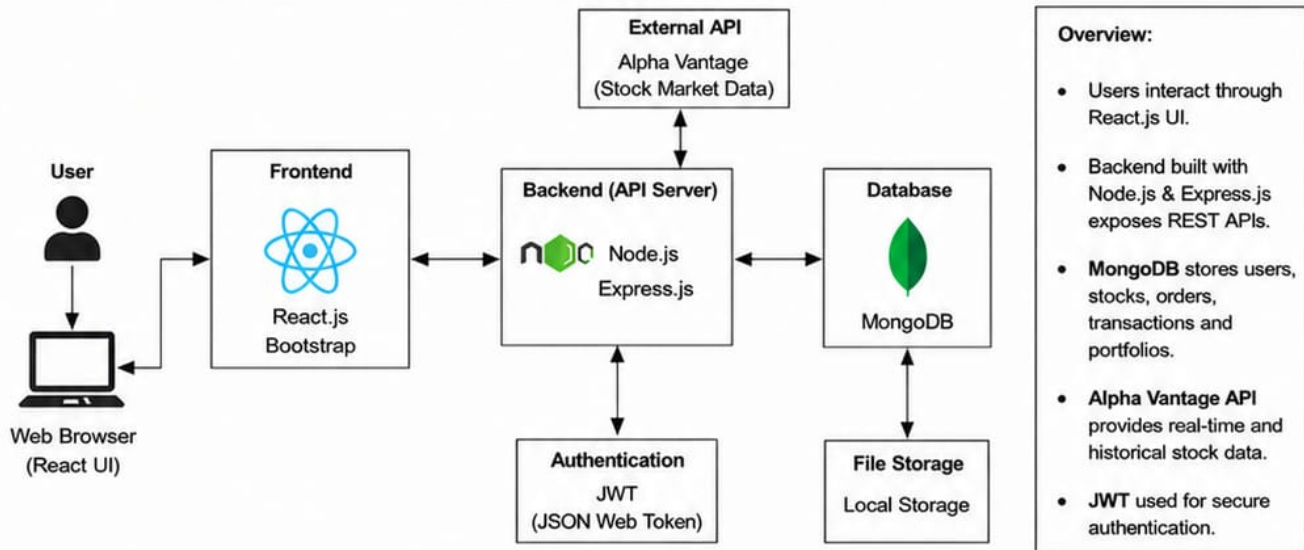


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface for users to browse stocks, trade, manage portfolio, track performance.	React.js, HTML5, CSS3, Bootstrap, JavaScript
2.	Application Logic-1	Server-side application, REST API handling business logic.	Node.js, Express.js
3.	Application Logic-2	Authentication & Authorization using JWT.	JSON Web Token (JWT)
4.	Application Logic-3	Fetching real-time and historical stock data.	Alpha Vantage API
5.	Database	Stores users, stocks, orders, transactions, portfolios.	MongoDB (NoSQL)
6.	Cloud Database	(Future) Cloud database service for scalability.	MongoDB Atlas
7.	File Storage	Store uploaded files (profile images, documents).	Local Storage / Cloudinary (Future)
8.	External API-1	Real-time stock prices & market data.	Alpha Vantage API
9.	External API-2	(Future) Payment gateway for real trades.	Razorpay / Stripe (Future)
10.	Machine Learning Model	(Future) Stock price prediction model.	Python (Scikit-learn / TensorFlow)
11.	Infrastructure (Server / Cloud)	Deployment on cloud platform.	Render / Railway / Heroku (Cloud)

Table-2: Application Characteristics:

S.No	Characteristic	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used.	React.js, Node.js, Express.js, MongoDB, Bootstrap
2.	Security Implementations	Security and access control mechanisms implemented.	JWT Authentication, Bcrypt (Password Hashing), HTTPS, CORS, Input Validation
3.	Scalable Architecture	Justify the scalability of architecture.	3-Tier Architecture (Client, Server, Database) API based design for scalability
4.	Availability	Justify the availability of application.	Cloud deployment, Auto-restart, Regular Backups, Uptime monitoring
5.	Performance	Design consideration for performance.	Indexed MongoDB collections, Efficient API responses, Caching (Future)